

01_WM.R

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```
### Willkommen in R!
```

```
# STEP 1
```

```
# Clean previous variables
```

```
rm(list=ls(all=TRUE))
```

```
# Load data
```

```
wage_data = read.table("wages1.txt", header=TRUE, sep="\t", na.strings="NA",  
dec=".", strip.white=TRUE)  
head(wage_data)
```

```
##   EXPER MALE SCHOOL      WAGE  
## 1      9     0      13 6.315296  
## 2     12     0      12 5.479770  
## 3     11     0      11 3.642170  
## 4      9     0      14 4.593337  
## 5      8     0      14 2.418157  
## 6      9     0      14 2.094058
```

```
summary(wage_data)
```

```
##      EXPER      MALE      SCHOOL      WAGE  
## Min.   : 1.000   Min.   :0.0000   Min.   : 3.00   Min.   : 0.07656  
## 1st Qu.: 7.000   1st Qu.:0.0000   1st Qu.:11.00   1st Qu.: 3.62157  
## Median : 8.000   Median :1.0000   Median :12.00   Median : 5.20578  
## Mean   : 8.043   Mean   :0.5237   Mean   :11.63   Mean   : 5.75759  
## 3rd Qu.: 9.000   3rd Qu.:1.0000   3rd Qu.:12.00   3rd Qu.: 7.30451  
## Max.   :18.000   Max.   :1.0000   Max.   :16.00   Max.   :39.80892
```

```
# STEP 2
```

```
#warm up yourself w dfs
```

```
#wage data is a data frame, ie various types of vars included
```

```
#get diemsnion of data frame
```

```
dim(wage_data)
```

```
## [1] 3294    4
```

```
#get slots
```

```
str(wage_data)
```

```
## 'data.frame':    3294 obs. of  4 variables:
```

```
## $ EXPER : int  9 12 11 9 8 9 8 10 12 7 ...
```

```
## $ MALE : int 0 0 0 0 0 0 0 0 0 0 ...
## $ SCHOOL: int 13 12 11 14 14 14 12 12 10 12 ...
## $ WAGE : num 6.32 5.48 3.64 4.59 2.42 ...
```

#see first rows in full

```
head(wage_data)
```

```
##   EXPER MALE SCHOOL    WAGE
## 1     9     0     13 6.315296
## 2    12     0     12 5.479770
## 3    11     0     11 3.642170
## 4     9     0     14 4.593337
## 5     8     0     14 2.418157
## 6     9     0     14 2.094058
```

#get a certain recordset, eg row 5 column 1

```
wage_data[5,1]
```

```
## [1] 8
```

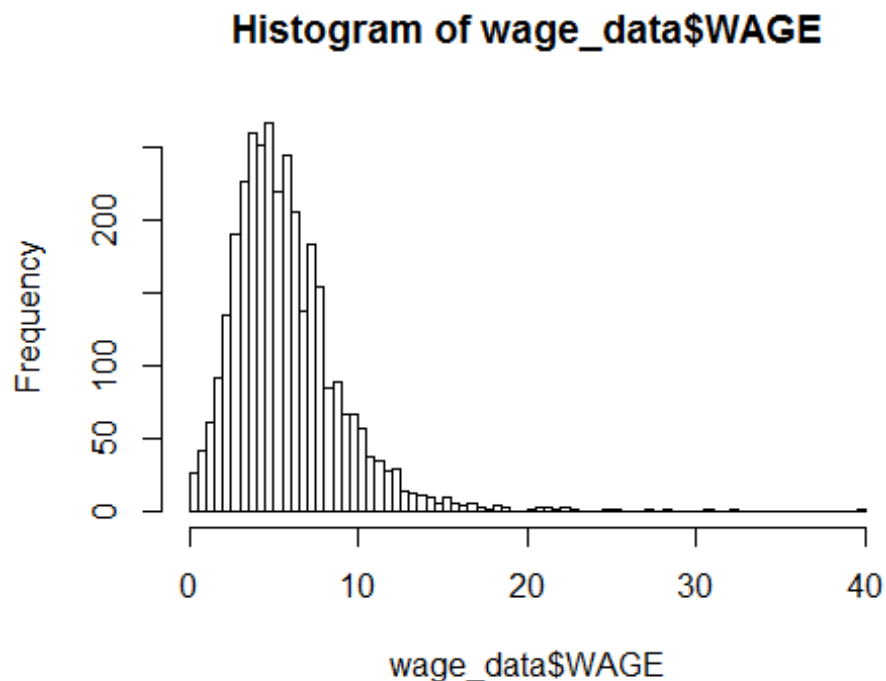
#get a certain column

```
head(wage_data$WAGE)
```

```
## [1] 6.315296 5.479770 3.642170 4.593337 2.418157 2.094058
```

Plot ad hoc charts

```
hist(wage_data$WAGE, nclass = 100)
```



STEP 3

```
# Calc mean, SD and show Boxplots
```

```
mean(wage_data$WAGE)
```

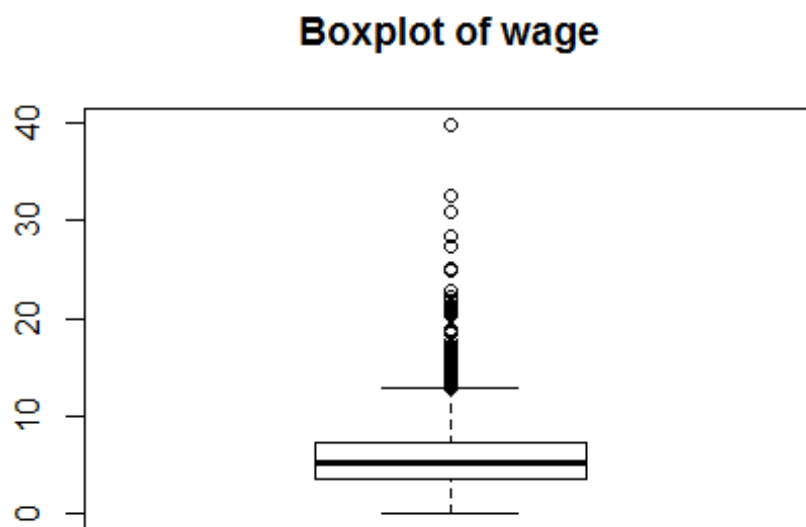
```
## [1] 5.757585
```

```
sd(wage_data$WAGE)
```

```
## [1] 3.269186
```

```
x11()
```

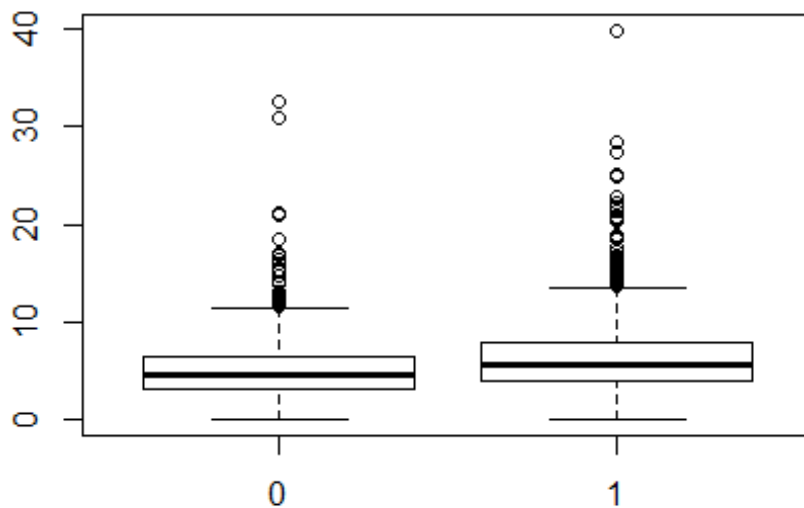
```
boxplot(wage_data$WAGE, main = "Boxplot of wage")
```



```
x11()
```

```
boxplot(wage_data$WAGE ~ wage_data$MALE, main = "Conditioned boxplots of wage")
```

Conditioned boxplots of wage



```
# STEP 4

#how many males?
no_males = sum(wage_data$MALE)
no_females = length(wage_data$MALE)-no_males
# cross chk
no_total = no_males + no_females

wages_males = rep(0,no_males)
wages_females = rep(0,no_females)

m = f = 1
for (i in (1:no_total)){
  if(wage_data$MALE[i] == 1)
  {
    wages_males[m] = wage_data$WAGE[i]
    m = m+1
  }
  else
  {
    wages_females[f] = wage_data$WAGE[i]
    f = f+1
  }
}

#some details of female wages
wfs = sort(wages_females)
head(wfs)
```

```
## [1] 0.07655561 0.09799118 0.10289074 0.21231423 0.22966683 0.27132808
```

```
tail(wfs)
```

```
## [1] 16.97183 18.51578 20.99811 21.23142 30.86217 32.49740
```

```
wilcox.test(wages_males,wages_females)
```

```
##
```

```
## Wilcoxon rank sum test with continuity correction
```

```
##
```

```
## data: wages_males and wages_females
```

```
## W = 1643200, p-value < 2.2e-16
```

```
## alternative hypothesis: true location shift is not equal to 0
```

```
#and now with a direction
```

```
#H0 males earn less than females?
```

```
wilcox.test(wages_males,wages_females,alternative = c("greater"))
```

```
##
```

```
## Wilcoxon rank sum test with continuity correction
```

```
##
```

```
## data: wages_males and wages_females
```

```
## W = 1643200, p-value < 2.2e-16
```

```
## alternative hypothesis: true location shift is greater than 0
```

```
#rather not ....
```

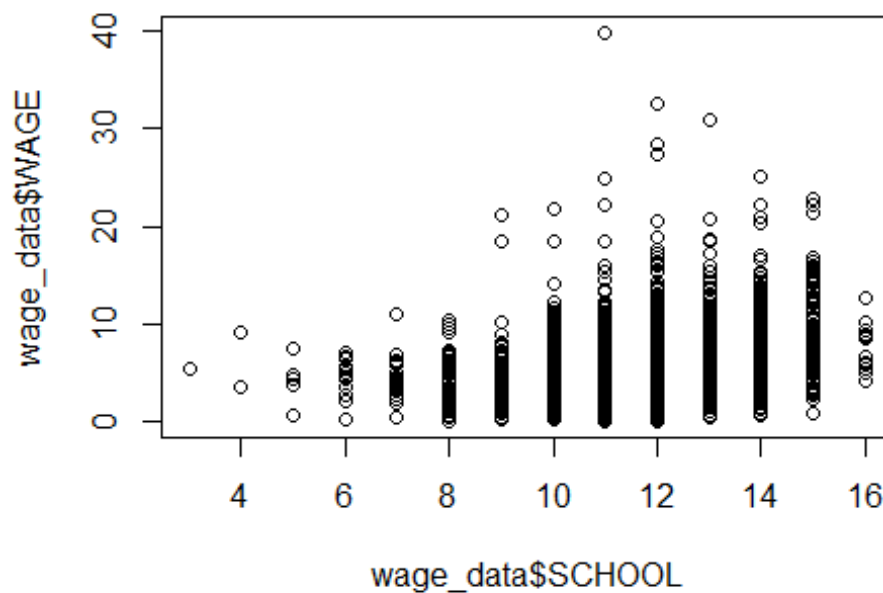
```
# STEP 5
```

```
cor(wage_data)
```

```
##          EXPER      MALE      SCHOOL      WAGE
## EXPER  1.00000000  0.1295449 -0.1935653  0.05138051
## MALE    0.12954487  1.0000000 -0.1190839  0.17817368
## SCHOOL -0.19356533 -0.1190839  1.0000000  0.28249249
## WAGE    0.05138051  0.1781737  0.2824925  1.00000000
```

```
x11()
```

```
plot(wage_data$WAGE ~ wage_data$SCHOOL)
```



```
#define above avg features
above_school = (wage_data$SCHOOL > mean(wage_data$SCHOOL))
above_wage = (wage_data$WAGE > mean(wage_data$WAGE))
table(above_school,above_wage)

##           above_wage
## above_school FALSE TRUE
##      FALSE    928  425
##      TRUE     977  964

chisq.test(table(above_school,above_wage))

##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  table(above_school, above_wage)
## X-squared = 108.18, df = 1, p-value < 2.2e-16

test = chisq.test(table(above_school,above_wage))
test$p.value

## [1] 2.453543e-25

test$statistic

## X-squared
## 108.1807

x11()#bonus stuff
mosaicplot(table(above_school,above_wage))
```

table(above_school, above_wage)

		FALSE	TRUE
above_wage	FALSE		
TRUE			
	above_school		

```
#cross check chi sq test
table(above_school,above_wage)

##           above_wage
## above_school FALSE TRUE
##      FALSE   928  425
##      TRUE    977  964

#put into RowsCols
RowsCols = table(above_school,above_wage)
#total sample size
TotalSample = sum(RowsCols)
TotalSample

## [1] 3294

#now go for eij as in Script WM 02 page 102
h2. = sum(RowsCols[2,]) # is h2., ie sum over 2nd row
#accordingly
h1. = sum(RowsCols[1,]) # is h1., ie sum over 1st row

h.1 = sum(RowsCols[,1])
h.2 = sum(RowsCols[,2])

#IF H0 that column and rows are independent THEN the expected probs in 2nd
row are
#eij = hi. * h.j / TotalSample , note that independence allows multiplication
of individual
#row and column probs, ie as usual to get joint prob if above shool and above
```

wage are independent

```
e11 = h1.*h.1 / TotalSample
e12 = h1.*h.2 / TotalSample
e22 = h2.*h.2 / TotalSample
e21 = h2.*h.1 / TotalSample
```

#test stat by feet

```
my_stat = (RowsCols[1,1]-e11)^2 / e11 + (RowsCols[1,2]-e12)^2 / e12
my_stat = my_stat + (RowsCols[2,1]-e21)^2 / e21 + (RowsCols[2,2]-e22)^2 / e22
```

#compare

```
my_stat
```

```
## [1] 108.9279
```

```
test$statistic
```

```
## X-squared
```

```
## 108.1807
```

STEP 6

```
reg = lm(wage_data$WAGE ~ wage_data$SCHOOL)
summary(reg)
```

```
##
```

```
## Call:
```

```
## lm(formula = wage_data$WAGE ~ wage_data$SCHOOL)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -6.744 -2.024 -0.482  1.443 34.403
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)   -0.72251    0.38739  -1.865   0.0623 .
```

```
## wage_data$SCHOOL  0.55716    0.03298  16.896  <2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

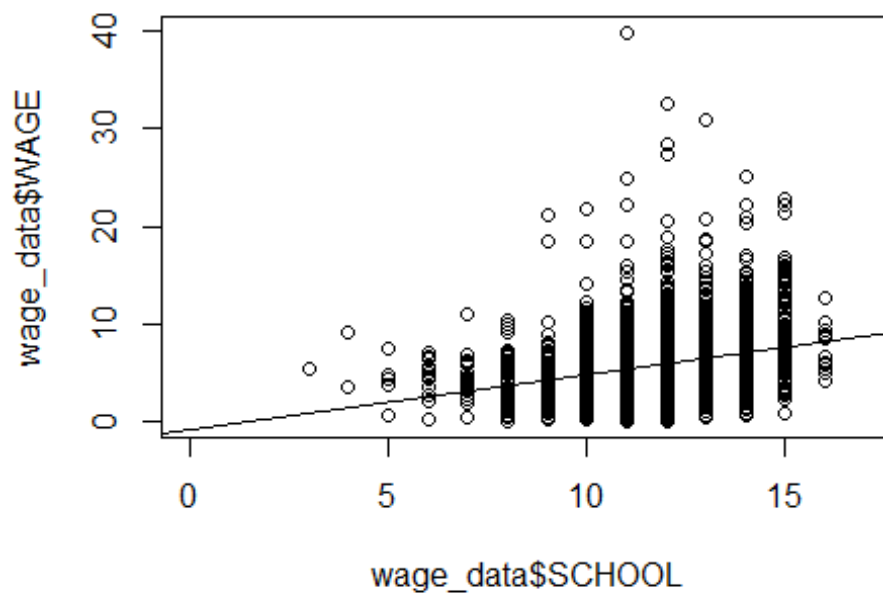
```
## Residual standard error: 3.137 on 3292 degrees of freedom
```

```
## Multiple R-squared:  0.0798, Adjusted R-squared:  0.07952
```

```
## F-statistic: 285.5 on 1 and 3292 DF, p-value: < 2.2e-16
```

```
plot(wage_data$WAGE ~ wage_data$SCHOOL, xlim = c(0,17))
```

```
abline(reg)
```



```
#cross check beta
coef(reg)[2]

## wage_data$SCHOOL
##      0.5571617

#by feet
cov(wage_data$WAGE ,wage_data$SCHOOL)/var(wage_data$SCHOOL)

## [1] 0.5571617

#transform into correlation
coef(reg)[2]*sd(wage_data$SCHOOL)/sd(wage_data$WAGE)

## wage_data$SCHOOL
##      0.2824925

#compare
cor(wage_data$SCHOOL,wage_data$WAGE)

## [1] 0.2824925

# STEP 7

#go from simple to multiple regression
regMult = lm(wage_data$WAGE ~ wage_data$SCHOOL + wage_data$MALE +
wage_data$EXPER)
summary(regMult)
```

```
##
## Call:
## lm(formula = wage_data$WAGE ~ wage_data$SCHOOL + wage_data$MALE +
##     wage_data$EXPER)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.654 -1.967 -0.457  1.444 34.194
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -3.38002    0.46498  -7.269 4.50e-13 ***
## wage_data$SCHOOL  0.63880    0.03280  19.478 < 2e-16 ***
## wage_data$MALE    1.34437    0.10768  12.485 < 2e-16 ***
## wage_data$EXPER    0.12483    0.02376   5.253 1.59e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.046 on 3290 degrees of freedom
## Multiple R-squared:  0.1326, Adjusted R-squared:  0.1318
## F-statistic: 167.6 on 3 and 3290 DF,  p-value: < 2.2e-16

#see that plain vanilla argument is weaker!
mean(wages_males)-mean(wages_females)

## [1] 1.166097

#end
```